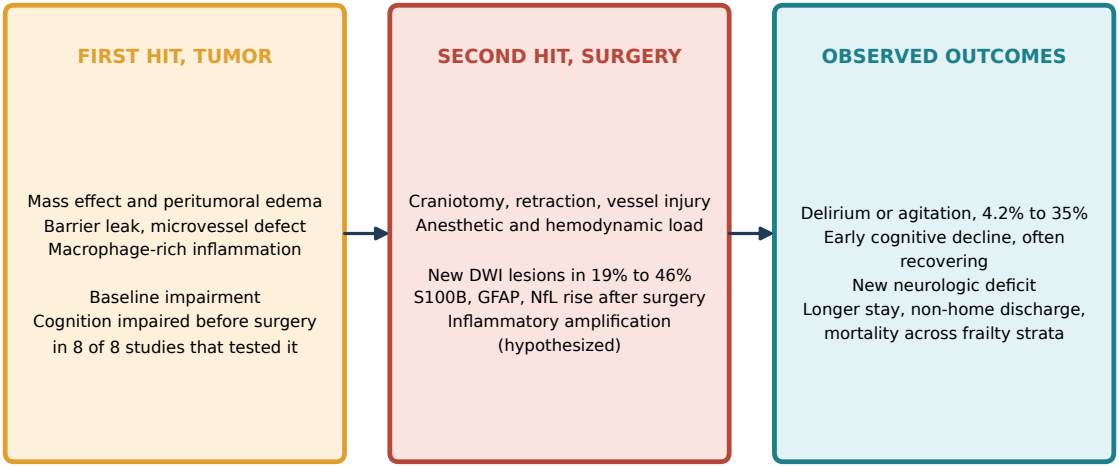
FIRST HIT, TUMOR
Mass effect and peritumoral edema
Barrier leak, microvessel defect
Macrophage-rich inflammation
Baseline impairment
Cognition impaired before surgery
in 8 of 8 studies that tested it
SECOND HIT, SURGERY
Craniotomy, retraction, vessel injury
Anesthetic and hemodynamic load
New DWI lesions in 19% to 46%
S100B, GFAP, NfL rise after surgery
Inflammatory amplification
(hypothesized)
OBSERVED OUTCOMES
Delirium or agitation, 4.2% to 35%
Early cognitive decline, often
recovering
New neurologic deficit
Longer stay, non-home discharge,
mortality across frailty strata